Median Index Preservation under Bounded Score Perturbations

Research Architecture Runtime

`operator/median`

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Abstract

The median index is the unique strict k-th order statistic for the conventional median rank k. When every pairwise score gap exceeds 2ε, that index is invariant under every additive perturbation with $\|\delta\|_\infty \leq \varepsilon$. The claim is Lean-certified (`LEAN_FULL`) as a definitional reduction of the k-th order-statistic margin theorem.

This theorem is: `Reduction`

1 Problem

The median operator on a finite score vector $s \in \mathbb{R}^n$ $(n \geq 2)$ returns the unique index $i^\star$ that is the strict k-th smallest coordinate of s, where k is the conventional median rank (so that exactly k scores are strictly smaller than $s_{i^\star}$, and no other index shares the value $s_{i^\star}$). The selected object is this index (equivalently, the unique median position under a strict-order presentation), not a set-valued median band.

2 Stability notion

Admissible perturbations are additive score noise $\delta \in \mathbb{R}^n$ in the closed ℓ_∞ ball $\|\delta\|_\infty \leq \varepsilon$ for a fixed $\varepsilon \geq 0$. Stability means: if $i^\star$ is the unique strict k-th smallest index of s and every pairwise gap of s exceeds 2ε, then $i^\star$ remains the unique strict k-th smallest index of $s + \delta$ for every such δ. Sharpness: if some rival index lies within distance 2ε of $s_{i^\star}$, an admissible δ destroys uniqueness of the strict k-th position.

3 Definitions

Definition 1 (Strict k-th smallest)**.** Fix $n \geq 2$, $s : \{0, \ldots, n-1\} \to \mathbb{R}$, and $k < n$. An index i is the *strict k-th smallest* if $\#\{j : s_j < s_i\} = k$ and $s_j = s_i$ forces $j = i$.

Definition 2 (All-gaps margin)**.** Scores satisfy $\mathrm{AllGapsExceed}(s, g)$ if $|s_p - s_q| > g$ whenever $p \neq q$.

Definition 3 (ℓ_∞ ball)**.** $\|\delta\|_\infty \leq \varepsilon$ means $|\delta_t| \leq \varepsilon$ for every coordinate t.

Assumptions. $n \geq 2$, $\varepsilon \geq 0$, a unique strict k-th smallest index exists, and (for invariance) $\mathrm{AllGapsExceed}(s, 2\varepsilon)$.

4 Theorem

Theorem 4 (Median / k-th margin invariance)**.** *Let $n \geq 2$, $\varepsilon \geq 0$, and let i be the strict k-th smallest index of s. If* $\mathrm{AllGapsExceed}(s, 2\varepsilon)$*, then for every δ with $\|\delta\|_\infty \leq \varepsilon$, the same index i is the strict k-th smallest index of $s + \delta$.*

Theorem 5 (Sharpness)**.** *Under the same standing hypotheses on (n, s, k, i) and $\varepsilon \geq 0$, if there exists $j \neq i$ with $|s_i - s_j| \leq 2\varepsilon$, then there exists δ with $\|\delta\|_\infty \leq \varepsilon$ such that i is* not *the strict k-th smallest index of $s + \delta$.*

5 Intuition

An ℓ_∞ adversary can decrease one coordinate by at most ε and increase another by at most ε, consuming at most 2ε of any pairwise gap. If every gap is strictly larger than 2ε, no pairwise order can reverse and no tie can form, so the count of scores below s_i and the uniqueness of s_i are preserved. If some gap is at most 2ε, a midpoint collision forces a tie and uniqueness fails.

6 Examples

Example 6. Take $n = 3$ scores $(s_0, s_1, s_2) = (1, 4, 7)$ and median rank $k = 1$ (unique middle value at index 1). Pairwise gaps are 3, 6, and 3, so $\mathrm{AllGapsExceed}(s, 2\varepsilon)$ holds whenever $2\varepsilon < 3$, e.g. $\varepsilon = 1$. For any $\|\delta\|_\infty \leq 1$, the middle index remains 1: the perturbed values stay strictly ordered. If instead $\varepsilon = 2$ so that $2\varepsilon = 4$ meets the gap $|4 - 1| = 3 \leq 4$, the midpoint adversary between indices 0 and 1 produces a tie and destroys uniqueness of the strict first order statistic.

7 Proof

Write $s' = s + \delta$ and assume $\|\delta\|_\infty \leq \varepsilon$.

Pairwise order preservation. Fix distinct indices p, q with $|s_p - s_q| > 2\varepsilon$. Without loss of generality suppose $s_p < s_q$, so $s_q - s_p > 2\varepsilon$. Then

$$s'_p - s'_q = (s_p - s_q) + (\delta_p - \delta_q) \leq -(s_q - s_p) + 2\varepsilon < 0,$$

because $|\delta_p - \delta_q| \le 2\varepsilon$. Hence $s'_p < s'_q$. The symmetric argument gives the converse direction: $s'_p < s'_q$ forces $s_p < s_q$. Thus

$$s_p < s_q \qquad \Longleftrightarrow \qquad s'_p < s'_q.$$

Count preservation. Applying the pairwise equivalence at $q = i$ yields $s_j < s_i \Leftrightarrow s'_j < s'_i$ for every j. Therefore

$$\#\{j : s_j < s_i\} = \#\{j : s'_j < s'_i\}.$$

If i is strict k-th smallest for s, the left-hand side equals k, so the right-hand side equals k.

Uniqueness preservation. Suppose $s'_j = s'_i$. If $j \ne i$, then $|s_j - s_i| > 2\varepsilon$ by the all-gaps hypothesis, and the same ℓ_∞ estimate used above yields $s'_j \ne s'_i$, a contradiction. Hence $j = i$.

Combining the two paragraphs shows i remains the strict k-th smallest index of s', which is Theorem **??**. (Median specialization: take the conventional median rank k; the argument never uses any property of k beyond $k < n$.)

Sharpness. Let $j \ne i$ satisfy $|s_i - s_j| \le 2\varepsilon$. Define the midpoint collision

$$\delta_t = \begin{cases} -(s_i - s_j)/2 & t = i, \\ (s_i - s_j)/2 & t = j, \\ 0 & \text{otherwise.} \end{cases}$$

Then $|\delta_t| \le \varepsilon$ for all t (since $|s_i - s_j|/2 \le \varepsilon$), and $s'_i = s'_j$. Uniqueness of a strict k-th value at i therefore fails, proving Theorem **??**.

8 Formal statement

Lean module

`Research.Operators.Median.Preservation`

Source file

`lean/Research/Operators/Median/Preservation.lean`

Theorems

- `median\_margin\_invariance`
- `median\_margin\_sharpness`

Note

Definitional reduction of OrderStat.KthMargin.

Certificate
`lean/certificates/median/median-margin`

Status
`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- Reduces to `Research.Operators.OrderStat.KthMargin` (`kth\_margin\_invariance`, `kth\_margin\_sharpness`).
- Uses the ℓ_∞ gap estimate $|(\delta_p - \delta_q)| \leq 2\varepsilon$ (triangle inequality for absolute values).
- Uses the definitions of strict k-th smallest, all-gaps margin, and midpoint collision adversary from that core.
- No other operator theorems are required.

10 Consequences

Median is the canonical specialization of unique strict order-statistic selection. The same margin theorem governs `quantile`, `percentile`, and `kth-order-statistic` (all reduce to the same k-th core). Ranking and top-k operators use the related pairwise ranking core rather than this count-based k-th statement; together they form the order-statistic layer of the primitive library. Any pipeline that selects a median index under bounded score noise may invoke Theorem **??** as the stability certificate.